Does an extract of carob (Ceratonia siliqua L.) have chemopreventive potential related to oxidative stress and drug metabolism in human colon cells?

. . . . .

Phenolic ingredients of an aqueous carob extract are well characterized and consist of mainly gallic acid (GA). In order to assess possible chemopreventive mechanisms of carob, which can be used as a cacao substitute, effects on expression of genes related to stress response and drug metabolism were studied using human colon cell lines of different transformation state (LT97 and HT29). Stress-related genes, namely catalase (CAT) and superoxide dismutase (SOD2), were induced by carob extract and GA in LT97 adenoma, but not in HT29 carcinoma cells. Although corresponding protein products and enzyme activities were not elevated, pretreatment with carob extract and GA for 24 h reduced DNA damage in cells challenged with hydrogen peroxide (H(2)O(2)). In conclusion, carob extract and its major phenolic ingredient GA modulate gene expression and protect colon adenoma cells from genotoxic impact of H(2)O(2). Upregulation of stress-response genes could not be related to functional consequences.